

LatentWire: Source-Private Byte-Scale Packet Communication with Destructive Controls

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Paper under double-blind review

Abstract

Multi-model language systems usually communicate by emitting text or by sharing internal state such as KV caches. Text is portable but exposes source outputs; cache sharing is information-rich but moves and exposes dense model state. We study a deliberately narrower interface: can a frozen source model send a byte-scale, source-private packet that helps a frozen receiver on public multiple-choice reasoning tasks without sending source text, hidden states, KV cache, source logits, or labels? We present LatentWire, a packet protocol and evaluation framework with explicit byte accounting, a source-private threat model, and destructive controls for source-choice, wrong-row, coordinate-shuffle, same-budget text, and cross-family shortcuts. On ARC-Challenge test, an 8-byte payload / 11-byte framed packet reaches 0.344 accuracy across 10 seeds, above target-only 0.265 and same-budget structured text 0.300; on OpenBookQA test, a 3-byte / 6-byte packet reaches 0.378 across 5 seeds, above target-only 0.276 and same-budget text 0.350. The decisive audit is also limiting: explicit source-index communication reaches 0.346 on ARC-Challenge and 0.378 on OpenBookQA, so the current positive is narrow source-candidate transfer, not evidence synthesis beyond the source or a method that beats selected-candidate communication. We also report that a Phi-3 cross-family source fails and that simple cached connector repairs do not close the gap. The workshop contribution is a reproducible packet-transfer result, a strict falsification protocol, and a systems boundary artifact: headline packets are 6–11 framed bytes, while dense KV/cache-transfer regimes begin hundreds to thousands of bytes per token under our accounting. We do not claim solved latent communication, C2C superiority, or native GPU latency/HBM/energy wins.

1 Introduction

Future language-model systems will often use more than one model: a cheap model may screen candidates, a specialist may inspect hard cases, and a stronger receiver may combine evidence. The simplest interface is text. Recent cache-level systems communicate through model internals, such as fused or selectively shared KV caches (Fu et al., 2026; Shi et al., 2026). These approaches are powerful, but they couple communication to large hidden-state objects and serving substrates.

This paper asks whether a much smaller interface can carry useful source-side candidate evidence. In our setting, both models see the same public question and answer choices. The source performs private computation over those choices. The only communicated object is a short byte packet; the receiver never sees source text, source logits, source hidden vectors, KV cache, or the answer key. This is a side-information problem in the spirit of distributed source coding (Slepian & Wolf, 1973; Wyner & Ziv, 1976): the receiver already has the public prompt and its own computation, so the packet should carry only the source evidence that is useful at the decoder. The scientific contribution is therefore not just a packet row, but a reviewable protocol for asking whether such rows survive the controls that usually create false positives.

Our strongest current result is deliberately scoped. On ARC-Challenge and OpenBookQA (Clark et al., 2018; Mihaylov et al., 2018), same-family Qwen2.5 packet trans-

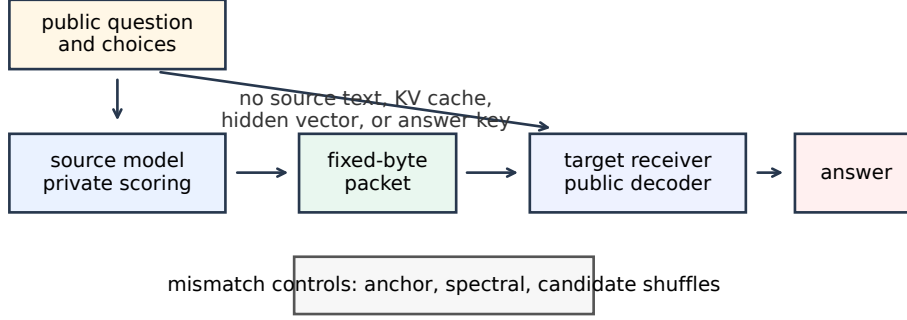


Figure 1: LatentWire protocol. Both sides receive the public task. The source privately scores candidates and emits only a fixed-byte packet. The receiver decodes the packet with a public basis and its own side information. Destructive controls scramble the shared coordinate system or candidate relation.

45 fer (Yang et al., 2025) improves over target-only and same-budget text controls. However, a
 46 strict Phi-3 source-family replacement (Abdin et al., 2024) fails, and additional connector
 47 repairs fail on the available TinyLlama/Qwen cached artifacts. This boundary matters: the
 48 paper is not claiming solved cross-family latent communication.

49 The contributions are:

- 50 • A byte-scale, source-private packet protocol and threat model for multiple-choice
 51 candidate transfer with no source text, hidden vector, KV cache, source-logit, or
 52 gold-label exposure.
- 53 • A strict evaluation suite that combines target-only, same-budget text, source-
 54 index/rank, coordinate-shuffle, wrong-row, same-source-choice, cross-family, and
 55 cached-connector controls.
- 56 • Controlled narrow positives on ARC-Challenge and OpenBookQA with seed re-
 57 peats and paired uncertainty, accompanied by an explicit source-index audit show-
 58 ing that the current effect is mostly source-choice preserving.
- 59 • A failure-boundary analysis showing that Phi-3 cross-family transfer and simple
 60 cached connector repairs fail, so broad latent-language and cross-family claims are
 61 unsupported.
- 62 • A systems accounting artifact comparing raw, framed, cacheline-rounded, and
 63 dense KV/cache byte regimes while separating measured packet objects from
 64 analytical byte floors and future native GPU measurements.

65 2 Source-private packet transfer

66 Let a public multiple-choice instance be $x = (q, c_1, \dots, c_m)$ with answer choices c_i . A source
 67 model S and receiver R both observe x , but only S computes the source-side private evidence
 68 $e_S(x)$. A packet encoder emits

$$z = E(e_S(x), x), \quad |z| \leq B \text{ bytes.}$$

69 The receiver predicts

$$\hat{y} = D_R(x, z),$$

70 where D_R may use public task features and the receiver’s own state, but may not access
 71 source text, source hidden states, source KV cache, source logits, or answer labels.

72 We evaluate a packet only if it beats three surfaces: target-only $D_R(x, \emptyset)$, a same-budget
 73 structured text control, and destructive controls that preserve superficial packet size but

Control family	What it tests	Current role
target-only / zero-source	target-cache or no-packet effects	required baseline
same-budget visible text	whether text at the same byte budget suffices	required baseline
source-index / rank / score	selected-candidate shortcuts	main claim boundary
wrong-row / row shuffle	row-specific source causality	destructive control
same-source-choice wrong-row	source-choice-only transfer	hardening control
anchor/bin/coordinate shuffle	shared-coordinate dependence	mechanism control
candidate	answer-order artifacts	destructive control
roll/derangement		
source-family substitution	cross-family generality	falsification control
cached connector repair	easy learned receiver rescue	negative diagnostic

Table 1: Control suite. The workshop claim requires packet utility to be interpreted through these controls; the current positive is explicitly bounded by source-index and source-family failures.

74 destroy the intended source-receiver relation. We also audit explicit source-index, source-
75 label-text, source-rank-code, and entropy-matched random-index baselines. We report seed
76 repeats and paired bootstrap lower bounds. This matters because a packet can be useful
77 while mostly preserving the source’s selected candidate; such a result is source-private
78 transfer, not proof of a shared latent language, reasoning beyond the source endpoint, or
79 compression beyond a one-byte selected-candidate channel.

80 **Threat model.** The packet channel is source-private but not source-oblivious. The sender
81 may privately compute over the public prompt and answer choices, then transmit only a
82 bounded byte string. The receiver may use that byte string, the public prompt, and its own
83 computation. The receiver may not observe source generations, chain-of-thought, hidden
84 states, KV cache, logits, score vectors, or labels. A method fails the current workshop claim
85 if its apparent gain is matched by a target-only cache effect, same-budget visible text, explicit
86 source-index/rank/score communication, wrong-row transfer, or answer leakage.

87 **Control suite.** Table 1 lists the reviewer-facing controls used to interpret the reported rows.
88 The suite is intentionally stronger than the headline claim: several controls are designed to
89 collapse methods that only copy the source’s preferred answer choice.

90 3 Public-coordinate packets

91 The current ARC-Challenge implementation uses a public train-anchor chart. For each
92 candidate, we compute a public relative-coordinate feature against a frozen set of training
93 anchors, then project that feature into a low-frequency orthonormal DCT-II spectral basis.
94 The packet is a sparse signed syndrome in this shared coordinate system. The receiver
95 scores candidate compatibility with the packet in the same public basis, combining the
96 packet evidence with receiver-side candidate information.

97 The design is related to relative representations (Moschella et al., 2023): anchors provide a
98 coordinate system less tied to raw hidden dimensions. Our use is narrower. We do not claim
99 zero-shot latent stitching or global representation alignment; we use a public basis to define
100 a tiny task packet. Random shared anchors work nearly as well as semantic anchors in
101 the current hashed ARC-Challenge implementation, so the correct interpretation is shared
102 coordinate agreement, not semantic anchor names.

103 OpenBookQA uses the same packet-evaluation contract at a smaller 3-byte budget. Across
104 both tasks, the source-choice cache is answer-key-forbidden: the source sees public questions
105 and choices, not labels. The packet is evaluated on frozen test slices after validation selection.

Benchmark	split	bytes	seeds	src	packet	target	text	pkt-src	pkt-text
ARC-Challenge	test, 1172	8/11	10	0.346	0.344	0.265	0.300	-0.008	+0.025
OpenBookQA	test, 500	3/6	5	0.378	0.378	0.276	0.350	-0.006	+0.000

Table 2: Main audit rows. “Bytes” reports payload/framed bytes. “Src” is explicit answer-key-forbidden source-index accuracy before packet decoding. “Text” is the same-budget structured text control. “Pkt-src” and “pkt-text” are minimum paired-bootstrap 95% lower bounds for packet minus source-index and packet minus same-budget text. Packet-minus-target lower bounds are +0.038 on ARC-Challenge and +0.046 on OpenBookQA.

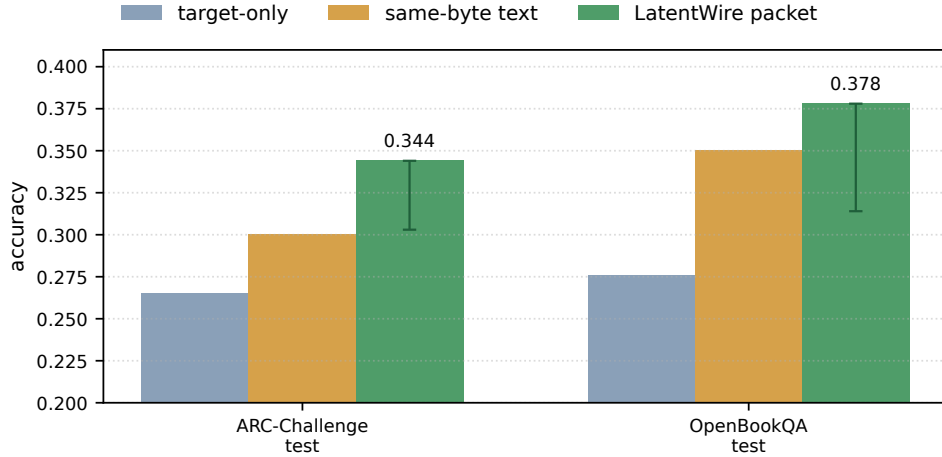


Figure 2: Packet accuracy against target-only and same-budget structured text controls. Error bars show the lower paired-bootstrap lift above target-only; upper uncertainty is omitted for readability because the gate is one-sided.

106 4 Main results

107 Table 2 and Figure 2 summarize the current positive evidence and the source-index audit.
 108 The ARC-Challenge row uses 10 packet/control seeds and 2000 paired bootstrap samples
 109 per seed on 1172 test examples. The OpenBookQA row uses 5 seeds on the 500-example
 110 test slice. The source-index audit is generated by reconstructing the same packet rows and
 111 adding explicit source-index, source-label-text, source-rank-code, and entropy-matched
 112 random-index baselines.

113 On ARC-Challenge, the matched Fourier/anchor-syndrome packet reaches 0.344 mean test
 114 accuracy with minimum seed accuracy 0.342. The answer-key-forbidden source index itself
 115 reaches 0.346, and packet predictions match the source-selected candidate at 0.995 mean
 116 rate across seeds. Anchor-ID shuffle, anchor-value shuffle, and spectral-bin permutation
 117 controls all fail in 0/10 seeds and collapse near target-only. The random shared-anchor
 118 diagnostic passes, which weakens any claim that named anchors are semantically special,
 119 but strengthens the claim that a shared public coordinate chart is the relevant mechanism.

120 On OpenBookQA, the 3-byte packet reaches 0.378–0.380 across seeds. The explicit source-
 121 index baseline reaches 0.378, and the packet follows the source index at 0.999 mean rate. The
 122 largest candidate-derangement control accuracy is 0.212, and the packet beats same-budget
 123 structured text in mean by +0.028, with a paired lower bound of +0.000. This second task
 124 reduces the risk that the result is only an ARC-Challenge artifact, but it confirms the same
 125 boundary: the current method is source-choice preserving and same-family, not cross-family
 126 latent reasoning.

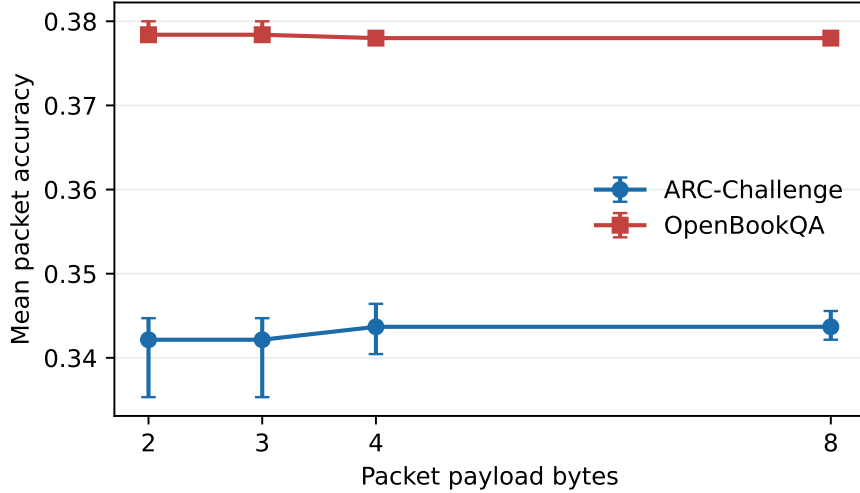


Figure 3: Payload-rate curve from the source-index audit. Packet payloads below 2 bytes are unsupported by the current sparse signed encoder; the 1-byte source-index baseline is reported separately in Table 2.

Source	surface	n	packet	target	text	ref. pkt	blocker
Phi-3	full test	1172	0.244	0.265	0.232	–	source quality
Phi-3	Qwen-disagree	833	0.200	0.273	0.203	0.340	source quality
TinyLlama	Qwen-disagree	473	0.269	0.268	0.258	0.317	family mismatch
Qwen2.5-1.5B	full test	1172	0.442	0.265	0.401	–	val incomplete

Table 3: Cross-family and source-family diagnostics. “Ref. pkt” is the Qwen-substituted packet on disagreement rows. Qwen2.5-1.5B is encouraging but same-family and validation-incomplete, so it is not a headline cross-family claim.

127 5 Falsification and failure decomposition

128 The most important negative result is the Phi-3 source-family replacement. We keep the
 129 same ARC-Challenge packet budget, public basis, seed count, and bootstrap budget, but
 130 replace the Qwen source-choice cache with a Phi-3 source. The packet no longer passes. On
 131 the full ARC-Challenge test slice, the Phi-3 packet reaches 0.244 versus target-only 0.265. On
 132 the stricter Qwen-disagreement slice, where the alternate source and Qwen source disagree,
 133 the Phi-3 packet reaches 0.200 versus 0.340 for the Qwen-substituted packet.

134 The failure decomposition separates three causes: source endpoint quality, packet corruption,
 135 and common-feature mismatch. For Phi-3, packet predictions follow the Phi-3 source choice
 136 at 0.997, but the source itself is below the target. For TinyLlama, packet fidelity is also
 137 high, but the source-family mismatch causes the disagreement-slice packet to lose to Qwen-
 138 substituted packets. For Qwen2.5-1.5B, the source is stronger and the packet is useful on
 139 frozen test, but the validation disagreement gate is incomplete. Across these rows, the
 140 packet mostly transports the source-selected candidate through the public coordinate basis.
 141 The source-index audit in Table 2 closes one obvious reviewer question: the current packet
 142 does not beat a direct selected-candidate code. The next research gate is therefore a richer
 143 common-feature connector or calibrated source-score packet that can improve beyond
 144 packet-only source choice.

145 We also tested cached learned repairs on the available local artifacts. A candidate-
 146 syndrome connector that sees packet and score-shape features reaches 0.288 versus 0.317
 147 for Qwen-substituted packets on frozen TinyLlama/Qwen disagreement rows. A one-

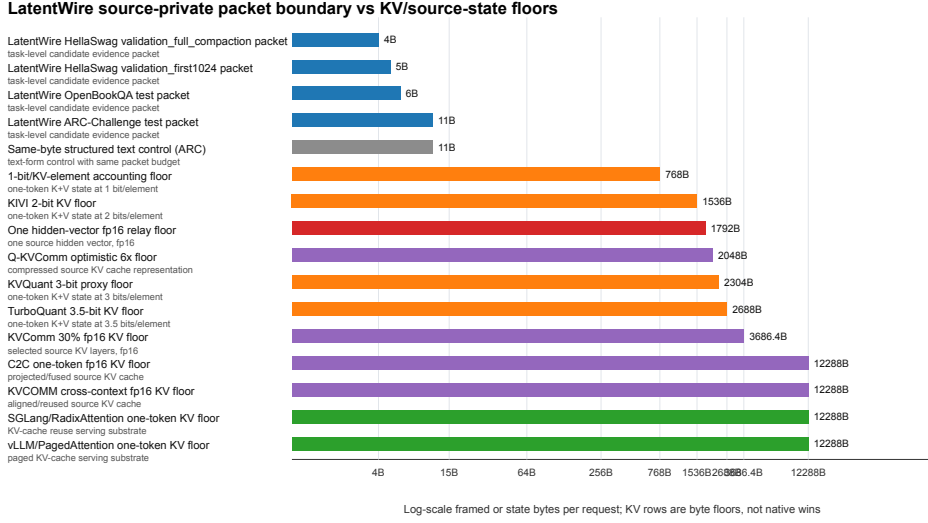


Figure 4: Boundary-byte accounting. Packet rows are measured task-packet objects. KV/cache rows are lower-bound floors or neighboring serving substrates, not native speed comparisons.

Method	raw	framed	text	KV/hidden	claim status
LatentWire OpenBookQA packet	3B	6B	no	no	local positive
LatentWire ARC-Challenge packet	8B	11B	no	no	local positive
Same-byte text control	8B	11B	yes	no	control only
1-bit/KV-element floor	768B	768B	no	KV	floor only
KIVI 2-bit KV floor	1536B	1536B	no	KV	floor only
C2C one-token KV floor	12288B	12288B	no	KV	native pending

Table 4: Exposure classes. Packet rows are measured/accounted local communication objects. KV/cache rows are analytical byte floors or native-pending baselines. The paper-ready systems claim is object size and source-exposure accounting, not GPU throughput, TTFT, TPOT, HBM, energy, or C2C superiority.

hidden-layer MLP over cached TinyLlama hidden/query summaries reaches 0.232, below Qwen-substituted packets, cached Tiny packets, target-only, and same-budget text. These negatives are useful: simple cached score geometry and mean hidden/query summaries do not explain the remaining oracle headroom.

6 Systems boundary

The systems contribution is byte/exposure accounting, not native serving throughput. Figure 4 compares framed packet bytes with one-token KV/cache-state accounting floors from neighboring systems and quantization methods. LatentWire sends 6–11 framed bytes on the two headline rows. A one-token 1-bit-per-KV-element accounting floor is 768 bytes in our model-configuration accounting; the largest headline packet is therefore 69.8x smaller than that floor. C2C-style cache transfer, KVComm-style selective sharing, and serving substrates such as vLLM/PagedAttention and SGLang operate on KV/cache objects rather than task-level packets (Fu et al., 2026; Shi et al., 2026; Kwon et al., 2023; Zheng et al., 2024). KV quantization work reduces cache bytes but still communicates or stores model state (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026).

Regime	Example	Object crossing interface	Difference from LatentWire
dense cache fusion	C2C	projected source KV cache	high-bandwidth, source-state exposing
selective KV sharing KV compression	KVComm KV quantizers	selected KV pairs/layers compressed KV/cache state	cache-level, native pending shrinks dense state, not source-private task packets
serving substrate	vLLM/SGLang	KV reuse and memory management	runtime baseline, not communication protocol
prompt/prefix compression	prefix/gist tokens	prompt-like tokens or parameters	not source-private model-to-model evidence
local shortcut base-lines	source-index/rank/score	selected answer or score sketch	must be beaten for stronger claims

Table 5: Related-work and baseline matrix. LatentWire is positioned as low-byte source-private packet transfer with destructive controls, not as a replacement for dense cache fusion or serving-runtime optimization.

7 Related work

Latent and relative representation communication. Relative representations use similarities to anchors to enable latent-space communication under invariances (Moschella et al., 2023). Sparse autoencoder work motivates interpretable feature coordinates (Huben et al., 2024), while feature-space universality work suggests that some features can be related across models (Lan et al., 2024). SAEbench provides evaluation machinery for sparse autoencoders (Karvonen et al., 2025), but these works do not by themselves provide a packet protocol or task-level evidence transfer result.

Prompt compression and learned prefixes. Prefix tuning and gist tokens learn continuous or discrete prompt-like interfaces within language models (Li & Liang, 2021; Mu et al., 2023). LatentWire differs because the communicated object is a byte packet decoded by a public receiver, not a learned soft prompt injected into one target. Query-bottleneck connectors such as BLIP-2’s Q-Former show that learned bottlenecks can bridge frozen model families across modalities (Li et al., 2023); our failed cached hidden/query summaries suggest that a true tokenwise query-bottleneck is the right future branch, but it is not closed here.

Cache communication and systems. Cache-to-cache and selective KV-sharing systems communicate richer internal state and are the closest competitors for model-to-model communication (Fu et al., 2026; Shi et al., 2026). vLLM and SGLang improve serving through KV-cache management and reuse (Kwon et al., 2023; Zheng et al., 2024). KV quantizers such as QJL, KIVI, KVQuant, and TurboQuant reduce state cost but remain state-transfer or state-storage techniques (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026). Table 5 summarizes how these baselines differ from LatentWire. Our current systems result is a boundary artifact against these state objects, not a native throughput win.

8 Limitations and next gate

The main limitation is scope. The positive results are same-family/source-cache results on public multiple-choice tasks, and the packet largely preserves the source’s selected candidate. We therefore do not claim to beat explicit source-index communication, source-score quantization, or calibrated source-choice baselines; those are required before a stronger compression claim. Phi-3 cross-family replacement fails, TinyLlama common-feature repair fails, and a true cross-family learned query-bottleneck remains future work. We also do not claim superiority over C2C, KVComm, QJL, KIVI, TurboQuant, vLLM, or SGLang in native serving metrics. Those rows require NVIDIA hardware and matched implementations.

The next exact gate for an ICLR-strength claim is a frozen-endpoint tokenwise connector with an explicit byte or query budget, tested against C2C/KV-style and structured-text baselines, with source-destroying controls and strict same-family versus cross-family separation. The current strongest claim is narrower: fixed-byte source-private packets can transfer

useful same-family evidence on ARC-Challenge and OpenBookQA, and the accompanying falsification ladder clarifies where the method stops working.

9 Conclusion

LatentWire shows that a tiny source-private candidate packet can improve frozen receiver decisions on two public QA benchmarks under strict same-family controls. The same evidence also draws a hard boundary: the current method is mostly source-choice preserving, is not a universal latent language, and does not yet solve cross-family communication. This combination of positive packet transfer, negative falsification, and systems byte/exposure accounting is the paper’s core.

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A Reviewer-facing claim audit

Table 6 states the claim boundary used to write the main paper. The full generated packet is in results/latentwire_colm.v3_review_packet_20260505/.

Claim	Status	Evidence	Required wording
source-private packet protocol	supported	packet rows and controls	protocol/evaluation contribution
narrow same-family packet utility	supported but narrow	ARC-Challenge and OpenBookQA rows	source-candidate transfer
shortcut collapses are common	supported	source-index and wrong-row audits	claim boundary, not failure-only story
beats C2C or dense KV transfer	unsupported	no native matched C2C row	do not claim
native GPU/HBM/energy win	unsupported	no native NVIDIA measurement	future runbook only
broad latent/cross-family communication	unsupported	Phi-3 and connector failures	ICLR future goal

Table 6: Claim audit used for COLM.v3 integration. Each main-text claim must map to one supported or supported-but-narrow row.

B Artifact manifest

- ARC-Challenge headline: colm_final/evidence/results/source_private_arc_challenge_fourier_anchor_syndrome_gate_20260502_budget8_10seed.b2000.
- Phi-3 falsification: colm_final/evidence/results/source_private_arc_challenge_source_family_cache_falsification_20260502_phi3_cpu_budget8_10seed.b2000.
- Cross-family decomposition: colm_final/evidence/results/source_private_arc_cross_family_failure_decomposition_20260502.
- Systems boundary: colm_final/evidence/results/source_private_systems_boundary_figure_table_20260502.
- OpenBookQA headline: colm_final/evidence/results/source_private_openbookqa_seed_stability_20260501_qwen05_hashed.test.3b.
- Source-index audit: colm_final/evidence/results/source_private_colm_acceptance_baselines_20260502.
- COLM.v3 review packet: results/latentwire_colm.v3_review_packet_20260505.

C Reproducibility summary

The artifact bundle records exact shell commands, input hashes, output hashes, and rerun results. The paper rows are produced from cached, answer-key-forbidden source-choice artifacts with the following settings:

Row	script	seeds	key settings
ARC-Challenge	Fourier/anchor syndrome gate	10	8B, 2000 bootstraps
OpenBookQA	seed-stability gate	5	3B, hashed features, 500 bootstraps
Source-index audit	acceptance baseline audit	10/5	1000 bootstraps, 2/3/4/8B rate curve
Phi-3	source-family cache falsification	10	8B, cached Phi-3 source
Systems	systems boundary table	–	byte/exposure accounting only

Frozen artifacts are hash-verified. In the final local pass, OpenBookQA, systems, and failure-decomposition reruns completed; the more expensive full ARC-Challenge/Phi-3 reruns are represented by frozen hashes unless rerun offline. Rerun JSON files are metric-oriented rather than byte-identical because they contain creation times and some artifacts include measured local latencies.

322 **Workshop checklist.** The current paper is ready for reviewer circulation only after the
323 generated COLM_v3 packet and the PDF agree on: (1) the main claim, (2) the source-private
324 threat model, (3) all main table values, (4) measured-versus-estimated systems labels, and
325 (5) unsupported claims. The repository command

326 `./venv_arm64/bin/python scripts/build_latentwire_colm_v3_review_packet.py`

327 regenerates the claim audit, table/figure inventory, systems ledger, artifact manifest, and
328 NVIDIA native runbook from tracked source artifacts.

329 **D Lay explanation of the main experiment**

330 The experiment asks whether one model can send a tiny hint to another model. Both models
331 see the same question and answer choices. The sender chooses a compressed coordinate hint,
332 only a few bytes long. The receiver uses that hint to choose an answer. If both sides agree on
333 the coordinate system, the hint helps. If we scramble the coordinate system, the hint stops
334 helping. That is evidence for a real shared packet interface on the tested same-family setting,
335 while the failed Phi-3 run shows the interface is not yet general across model families.